Mannadithya Aramudhan

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PROFILE

I am currently pursuing my M.Tech in CSE (AI & ML) at Vellore Institute of Technology, Vellore. Previously, I majored in Electronics and Communication Engineering. With a strong passion for continuous learning and problem-solving, I aspire to build a career in AI Engineering.

SKILLS

Programming: Python
Libraries & Frameworks: OpenCV, PyTorch, TensorFlow, spaCy, EasyOCR, PaddleOCR, NumPy, Pandas
Tools & Platforms: Streamlit, Neo4j, MS SQL Server, Git, Jupyter Notebook
Domains: Computer Vision, Natural Language Processing (NLP), OCR Systems, Deep Learning, Knowledge Graphs, BiLSTM,CNN, Model Training and Evaluation, Bias and Fairness Analysis, Explainable AI.
Soft Skills: Analytical Thinking, Problem-Solving, Research Orientation, Strong Communication, Collaboration, Adaptability

EDUCATION

Master of Technology - Computer Science and Engineering (Artificial Intelligence and Machine Learning)-VIT **[Vellore Institute of Technology]** (2024-2026)

Bachelor of Engineering – Electronics and Communication Engineering

[SRM Easwari Engineering College](2020-2024)

EXPERIENCE

AI Intern – L&T Constructions
(Ongoing)

- Engineered a 7-stage image enhancement pipeline (grayscale, border removal, deskew, denoise, brightness normalization, sharpening, adaptive binarization) using OpenCV, optimized for low-quality scanned invoices.
- Solved critical preprocessing challenges by removing scanner artifacts via morphological operations, correcting skew using Hough line detection (±15°), and eliminating noise using Non-Local Means denoising, significantly improving OCR readability.
- Tuned adaptive binarization (block Size=51, C=10) to preserve faint text while avoiding halo artifacts, improving extraction accuracy on degraded documents.
- Integrated PaddleOCR with regex-based parsing to extract structured fields such as GSTIN, invoice number, dates, quantities, and units from invoice images.
- Built an automated data pipeline using Neo4j and MS SQL to construct a knowledge graph linking suppliers, invoices, and line items for real-time tracking and anomaly detection.
- Developed an end-to-end Streamlit application supporting image/PDF uploads and real-time structured JSON output for non-technical users.

Project Intern – LTIMindtree
(August 2025 – October 2025)

- Currently developing a **Financial Clause Extractor, Risk Detector, and Explainer** — a complete **Text Extraction and NLP-based offline model** for analyzing financial documents.

- Implementing an end-to-end pipeline using **spaCy** for text segmentation and entity recognition, **EasyOCR** for clause extraction from scanned documents, and **Streamlit** for model deployment and interactive visualization.
- Designing a **risk scoring and explainability module** to identify, categorize, and interpret high-risk clauses within contracts.
- Ensuring offline operability and data privacy compliance under restricted compute environments.
- Performing iterative testing and evaluation for clause-level accuracy, interpretability, and user experience optimization.

Intern - Enthu Technologies
(Jun 2023-Feb 2024)

- IoT (Internet of Things) technology integrates a diverse range of sensors for data collection across various domains.
- My hands-on experience includes integrating a diverse range of sensors and leveraging the insights gathered to develop robust and scalable IoT solutions
- Including temperature, humidity, motion, proximity, light, acceleration, orientation, pressure, gas and chemical detecting sensor.

PROJECTS

Financial Clause Extractor, Risk Detector, and Explainer

- Developing an **offline NLP-based model** to analyze and interpret clauses in financial documents.
- Utilized **EasyOCR** for text extraction from scanned documents and **spaCy** for clause segmentation and entity recognition.
- Built a **risk detection and explainability module** for identifying and classifying high-risk clauses.
- Designed a **Streamlit dashboard** for visualization and interactive exploration of clause-level risks.
- Ensured **data privacy and offline operability** under restricted computing environments.

ESC-Net: Enhanced SAR Image Colorization Network – *VIT, Vellore*

- Developed a deep learning-based framework (ESC-Net) for **automatic colorization of grayscale Synthetic Aperture Radar (SAR) images**.
- Implemented **C-GAN-based architecture** with **U-Net generators** and **attention modules** to preserve structural and spectral details.
- Evaluated performance using **SSIM, PSNR, color fidelity, and detail preservation metrics**, achieving superior results compared to existing methods.

Image Transmission using Universal Filtered Multicarrier Modulation in Underwater Wireless Optical Communication (UWOC)

- Designed and implemented image transmission using universal filtered multicarrier modulation in underwater wireless optical communication
- Technologies like MatLab and LiFi module were used.
- Improved data rate and reliability of underwater communication under changing channel conditions.

Certificates:

https://drive.google.com/drive/folders/1maXkN1UhnlhfjaxzXCyi7J8xbExiUilo?usp=drive_link